The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Minimal Completion-State Shell-Trace Core
Necessity / Uniqueness from Completion-Balance Ambient Dynamics

Abstract

The current active theorem on the primitive-reservoir observer-reduction line has already sharpened the live internal burden below the completion-core frontier: the full minimal zero-response affine completion-core presentation $\mathfrak{C}^{\mathrm{zr,aff}}$ carries no extra benchmark-facing freedom beyond a smaller completion-state shell-trace core $\mathfrak{C}^{\mathrm{tr}}$. The remaining honest benchmark-facing move is therefore no longer to restate the larger completion-core graph package, but to derive or uniquely force that smaller trace core from still more primitive explicit substrate structure, or else collapse it further.

The present note takes the preferred first route. For each sector it introduces the same completion-balance ambient dynamics package already preserved on the active line: a compact convex ambient admissible body, affine observer-data and shell-response readouts, an invariant affine zero-hidden slice, a positive ambient balance operator whose kernel is exactly the zero-hidden translation space, exact splitting of the ambient body over zero-hidden and hidden directions, symmetry and time-reversal actions preserving the whole package, an exact zero-response shell realization, fixed-data shell solvability on the zero-hidden slice, and a coercive balance inequality on data-invisible zero-hidden directions modulo the inert completion kernel. From that package it derives directly the current live datum $\mathfrak{C}^{\mathrm{tr}}$: the faithful completion-state quotient of the zero-hidden slice, its compact convex state body, the descended affine observer-data and shell-response trace maps, the hidden body with positive gap operator, the induced symmetry and time-reversal structure, the exact zero-response trace realization, fixed-data solvability, and trace-kernel coercive control.

This is a direct forcing theorem for the current live burden rather than another restatement of a preserved deeper relay. The shell-coordinate construction yielding the current exact-shell affine nucleus $\mathfrak{N}^{\mathrm{aff}}$ already factors through the ambient-induced trace core, so the trace core is no longer an independently unexplained stopping point on the active line. The benchmark package remains fixed throughout: the operative metric is still exactly $g^{\sharp}$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has again tightened what counts as an honest next step. The direct exact-shell-affine-nucleus forcing theorem already removed $\mathfrak{N}^{\mathrm{aff}}$ as the live unexplained stopping point by deriving it canonically from the explicit completion-core package $\mathfrak{C}^{\mathrm{zr,aff}}$. The later trace-core collapse theorem then sharpened that burden once more by showing that the full affine completion-plane and completion-body presentation carries no extra benchmark-facing freedom beyond the smaller completion-state shell-trace core $\mathfrak{C}^{\mathrm{tr}}$.

That sharper routing result changes what a deeper-origin theorem must now accomplish. It is no longer enough to reopen a preserved deeper relay if the manuscript merely reproduces the same downstream output without addressing the smaller live datum itself. The present note therefore does not return to the ambient package as if the older completion-core theorem still fixed the live burden by itself. Instead it targets the current burden directly: it proves that the smaller trace core $\mathfrak{C}^{\mathrm{tr}}$ is itself a canonical and necessary reduction of completion-balance ambient dynamics.

This is the cleanest honest way to spend the preferred option left open by the current routing note. The theorem does not enlarge the benchmark package, does not introduce a second operative metric, and does not upgrade adoption to completed first-principles derivation. Its narrower benchmark-facing role is only to show that the present trace-core burden is already forced by a more primitive explicit substrate package. Once that is done, the burden moves below the trace core itself.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It proves only that the current live trace-core burden is a canonical output of a deeper completion-balance ambient dynamics package. It does not claim that the ambient package has already been derived from ultimate substrate microphysics, and it does not license stronger first-principles promotion language.

2. Fixed Primitive Continuation Data

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Remark 1. *Definition 1 is not reopened here. The primitive shell data, the one operative metric, and the ordinary matter route remain fixed continuation data. The present note only settles the current benchmark-facing burden below the trace-core frontier.*

3. Current Live Burden: Minimal Completion-State Shell-Trace Core

Definition 2 (Completion-state shell-trace core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* minimal completion-state shell-trace core with hidden-sector gap *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean completion-state space*

$$\mathcal{C}_\sigma,$$

the observer-data space

$$\mathcal{Y}^{\mathrm{obs}}_\sigma \equiv \widehat{X}_\sigma \times Z^{\mathrm{cur}}_\sigma \times Z^{\mathrm{eq}}_\sigma,$$

the shell-response space

$$E^{\mathrm{amb}}_\sigma,$$

and a hidden space

$$H_\sigma.$$

2. *A compact convex completion-state body*

$$P^{\mathrm{tr}}_\sigma \subset \mathcal{C}_\sigma$$

with nonempty interior in $\mathcal{C}_\sigma$.

3. *A compact convex hidden body*

$$D_\sigma \subset H_\sigma$$

with $0 \in \operatorname{int} D_\sigma$, *together with a positive self-adjoint hidden-sector operator*

$$K^{\mathrm{hid}}_\sigma : H_\sigma \to H_\sigma$$

satisfying

$$\left\langle \eta, K^{\mathrm{hid}}_\sigma \eta \right\rangle \geq \kappa^{\mathrm{hid}}_\sigma \|\eta\|^2 \qquad \textit{for all } \eta \in H_\sigma. \tag{2}$$

4. *Affine observer-data and shell-response maps*

$$\Omega^{\mathrm{tr,dat}}_\sigma : \mathcal{C}_\sigma \to \mathcal{Y}^{\mathrm{obs}}_\sigma, \qquad \Omega^{\mathrm{tr,sh}}_\sigma : \mathcal{C}_\sigma \to E^{\mathrm{amb}}_\sigma.$$

5. *A compact symmetry group*

$$\mathbb{K}_\sigma^{\mathrm{tr}} \subset O(\mathcal{C}_\sigma) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}}) \times O(H_\sigma)$$

and an orthogonal involution

$$\tau_\sigma^{\mathrm{tr}} \in O(\mathcal{C}_\sigma) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}}) \times O(H_\sigma)$$

preserving P_σ^{tr}, the exact zero-response trace slice

$$P_\sigma^{\mathrm{tr}} \cap \big(\Omega_\sigma^{\mathrm{tr,sh}}\big)^{-1}(0),$$

the hidden body D_σ, and K_σ^{hid}, and satisfying

$$\Omega_\sigma^{\mathrm{tr,dat}}(g_{\mathrm{st}} m) = g_{\mathrm{dat}} \Omega_\sigma^{\mathrm{tr,dat}}(m), \qquad \Omega_\sigma^{\mathrm{tr,sh}}(g_{\mathrm{st}} m) = g_{\mathrm{sh}} \Omega_\sigma^{\mathrm{tr,sh}}(m)$$

for every $m \in \mathcal{C}_\sigma$ and every

$$(g_{\mathrm{st}}, g_{\mathrm{dat}}, g_{\mathrm{sh}}, g_{\mathrm{hid}}) \in \mathbb{K}_\sigma^{\mathrm{tr}},$$

with the analogous intertwining equalities for the components of $\tau_\sigma^{\mathrm{tr}}$.

6. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth exact zero-response trace realization

$$\zeta_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{C}_\sigma$$

whose image is exactly

$$P_\sigma^{\mathrm{tr}} \cap \big(\Omega_\sigma^{\mathrm{tr,sh}}\big)^{-1}(0),$$

and such that

$$\Omega_\sigma^{\mathrm{tr,dat}}\big(\zeta_\sigma(w)\big) = \big(\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)\big) \tag{3}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$.

7. *Fixed-data solvability on the trace body:*

$$\Omega_\sigma^{\mathrm{tr,dat}}\big(P_\sigma^{\mathrm{tr}}\big) = \Omega_\sigma^{\mathrm{tr,dat}}\Big(P_\sigma^{\mathrm{tr}} \cap \big(\Omega_\sigma^{\mathrm{tr,sh}}\big)^{-1}(0)\Big). \tag{4}$$

8. *Trace-kernel coercivity: there exists $\kappa_\sigma^{\mathrm{tr}} > 0$ such that for every*

$$\delta m \in \mathcal{C}_\sigma$$

satisfying

$$D\Omega_\sigma^{\mathrm{tr,dat}}\, \delta m = 0,$$

one has

$$\big\| D\Omega_\sigma^{\mathrm{tr,sh}}\, \delta m \big\|^2 \geq \kappa_\sigma^{\mathrm{tr}} \|\delta m\|^2. \tag{5}$$

Remark 2. *Definition 2 is the current live internal burden on the active line. The present note*

asks whether those trace-level data are themselves forced by a still more primitive explicit substrate package, without reintroducing the larger completion-core graph presentation as the stopping point.

4. Deeper Explicit Package: Completion-Balance Ambient Dynamics

Definition 3 (Completion-balance ambient dynamics with inert completion kernel)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* completion-balance ambient dynamics package *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean ambient state space*

$$\mathcal{Z}_\sigma,$$

the observer-data space

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \equiv \widehat{X}_\sigma \times Z_\sigma^{\mathrm{cur}} \times Z_\sigma^{\mathrm{eq}},$$

an ambient shell-response space

$$E_\sigma^{\mathrm{amb}},$$

and a compact convex ambient admissible body

$$\mathcal{B}_\sigma^{\mathrm{bal}} \subset \mathcal{Z}_\sigma$$

with nonempty interior.

2. *An affine observer-data readout*

$$\Lambda_\sigma^{\mathrm{dat}} : \mathcal{Z}_\sigma \to \mathcal{Y}_\sigma^{\mathrm{obs}}$$

and an affine shell-response readout

$$\Lambda_\sigma^{\mathrm{sh}} : \mathcal{Z}_\sigma \to E_\sigma^{\mathrm{amb}}.$$

3. *An affine zero-hidden slice*

$$\mathcal{Z}_\sigma^0 \subset \mathcal{Z}_\sigma$$

with translation space

$$V_\sigma^0 \equiv T_z \mathcal{Z}_\sigma^0 \qquad (z \in \mathcal{Z}_\sigma^0),$$

and hidden complement

$$H_\sigma \equiv V_\sigma^{0\perp} \subset \mathcal{Z}_\sigma.$$

4. *A positive self-adjoint ambient balance operator*

$$\mathcal{K}_\sigma^{\mathrm{bal}} : \mathcal{Z}_\sigma \to \mathcal{Z}_\sigma$$

whose kernel is exactly the zero-hidden translation space,

$$\ker \mathcal{K}_\sigma^{\mathrm{bal}} = V_\sigma^0,$$

and whose restriction to the hidden complement obeys

$$\left\langle \eta, \mathcal{K}_\sigma^{\mathrm{bal}} \eta \right\rangle \geq \kappa_\sigma^{\mathrm{hid}} \|\eta\|^2 \qquad \textit{for all } \eta \in H_\sigma \tag{6}$$

for some $\kappa_\sigma^{\mathrm{hid}} > 0$.

5. *The hidden projection body*

$$D_\sigma \equiv \left\{ \eta \in H_\sigma : z + \eta \in \mathcal{B}_\sigma^{\mathrm{bal}} \textit{ for some } z \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \right\}$$

is compact and convex, satisfies

$$0 \in \operatorname{int} D_\sigma,$$

and the ambient admissible body splits exactly as

$$\mathcal{B}_\sigma^{\mathrm{bal}} = \left\{ z + \eta : z \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0, \quad \eta \in D_\sigma \right\}. \tag{7}$$

6. *A compact symmetry group*

$$\mathbb{B}_\sigma \subset O(\mathcal{Z}_\sigma) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

and an orthogonal involution

$$\vartheta_\sigma^{\mathrm{bal}} \in O(\mathcal{Z}_\sigma) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

preserving $\mathcal{B}_\sigma^{\mathrm{bal}}$, $\mathcal{Z}_\sigma^0$, $\Lambda_\sigma^{\mathrm{dat}}$, $\Lambda_\sigma^{\mathrm{sh}}$, *and* $\mathcal{K}_\sigma^{\mathrm{bal}}$.

7. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth zero-response shell realization

$$\zeta_\sigma^{\mathrm{bal}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{Z}_\sigma$$

whose image is exactly

$$\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \cap \left(\Lambda_\sigma^{\mathrm{sh}}\right)^{-1}(0),$$

and such that

$$\Lambda_\sigma^{\mathrm{dat}}(\zeta_\sigma^{\mathrm{bal}}(w)) = \left(\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)\right) \tag{8}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$.

8. *Fixed-data shell solvability on the zero-hidden slice:*

$$\Lambda_\sigma^{\mathrm{dat}}(\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0) = \Lambda_\sigma^{\mathrm{dat}}\Big(\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \cap \left(\Lambda_\sigma^{\mathrm{sh}}\right)^{-1}(0)\Big). \tag{9}$$

9. *The inert completion kernel*

$$K_\sigma^0 \equiv \ker\big(D\Lambda_\sigma^{\mathrm{dat}}|_{V_\sigma^0}\big) \cap \ker\big(D\Lambda_\sigma^{\mathrm{sh}}|_{V_\sigma^0}\big)$$

and the orthogonal completion-state projector

$$P_\sigma^{\mathrm{co}} : V_\sigma^0 \to K_\sigma^{0\perp} \cap V_\sigma^0.$$

10. *Completion-balance coercivity on data-invisible zero-hidden directions: there exists* $\kappa_\sigma^{\mathrm{co}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta z \in T_{\zeta_\sigma^{\mathrm{bal}}(w)} \mathcal{Z}_\sigma^0 = V_\sigma^0$$

satisfying

$$D\Lambda_\sigma^{\mathrm{dat}}\, \delta z = 0,$$

one has

$$\|D\Lambda_\sigma^{\mathrm{sh}}\, \delta z\|^2 \geq \kappa_\sigma^{\mathrm{co}}\, \|P_\sigma^{\mathrm{co}} \delta z\|^2. \tag{10}$$

Remark 3. *Definition 3 is deeper than the trace core in exactly the way needed now. It begins with an ambient admissible body, an invariant zero-hidden slice, and an inert kernel on that slice. The current live trace datum must then be recovered as the faithful zero-hidden quotient together with the descended ambient readouts.*

5. Canonical Trace Core Induced by Ambient Dynamics

Definition 4 (Canonical trace core induced by ambient dynamics)**.** *Assume Definition 3. Choose once and for all a reference shell point*

$$w_\sigma^* \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad z_\sigma^* \equiv \zeta_\sigma^{\mathrm{bal}}(w_\sigma^*).$$

Define the canonical trace core as follows.

1. *The completion-state space is*

$$\mathcal{C}_\sigma \equiv K_\sigma^{0\perp} \cap V_\sigma^0.$$

2. *The zero-hidden trace coordinate is the affine map*

$$\pi_\sigma^{\mathrm{tr}}(z) \equiv P_\sigma^{\mathrm{co}}(z - z_\sigma^*), \qquad z \in \mathcal{Z}_\sigma^0. \tag{11}$$

3. *The completion-state trace body is*

$$P_\sigma^{\mathrm{tr}} \equiv \left\{ \pi_\sigma^{\mathrm{tr}}(z) : z \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \right\}. \tag{12}$$

4. *The affine observer-data and shell-response trace maps are*

$$\Omega_\sigma^{\mathrm{tr,dat}}(m) \equiv \Lambda_\sigma^{\mathrm{dat}}(z_\sigma^* + m), \qquad \Omega_\sigma^{\mathrm{tr,sh}}(m) \equiv \Lambda_\sigma^{\mathrm{sh}}(z_\sigma^* + m) \tag{13}$$

for $m \in \mathcal{C}_\sigma$.

5. *The hidden operator is the restriction*

$$K_\sigma^{\mathrm{hid}} \equiv \mathcal{K}_\sigma^{\mathrm{bal}}\big|_{H_\sigma}. \tag{14}$$

6. *The induced symmetry group and time reversal are obtained by restricting the ambient $\mathcal{Z}_\sigma$-action to $\mathcal{C}_\sigma \oplus H_\sigma$ and keeping the recorded data and shell actions:*

$$\mathbb{K}_\sigma^{\mathrm{tr}} \equiv \{(g_{\mathrm{tr}}, g_{\mathrm{dat}}, g_{\mathrm{sh}}, g_{\mathrm{hid}}) : (g_{\mathrm{amb}}, g_{\mathrm{dat}}, g_{\mathrm{sh}}) \in \mathbb{B}_\sigma\},$$

where g_{tr} and g_{hid} are the restrictions of g_{amb} to $\mathcal{C}_\sigma$ and H_σ, and

$$\tau_\sigma^{\mathrm{tr}} \equiv (\vartheta_\sigma^{\mathrm{bal}}|_{\mathcal{C}_\sigma}, \vartheta_\sigma^{\mathrm{bal}}|_{\mathcal{Y}_\sigma^{\mathrm{obs}}}, \vartheta_\sigma^{\mathrm{bal}}|_{E_\sigma^{\mathrm{amb}}}, \vartheta_\sigma^{\mathrm{bal}}|_{H_\sigma}).$$

7. *The exact zero-response trace realization is*

$$\zeta_\sigma(w) \equiv \pi_\sigma^{\mathrm{tr}}(\zeta_\sigma^{\mathrm{bal}}(w)). \tag{15}$$

Remark 4. *The chosen reference shell point z_σ^* fixes only the affine origin of the completion-state coordinate. Changing it translates the trace coordinate on the same faithful quotient and does not alter any benchmark-relevant constituent.*

Proposition 1 (Ambient dynamics induce a trace core)**.** *Assume Definitions 3 and 4. Then the canonical data of Definition 4 satisfy every requirement of Definition 2.*

Proof. Because $K_\sigma^0 \subset V_\sigma^0$ and $\mathcal{C}_\sigma = K_\sigma^{0\perp} \cap V_\sigma^0$, every $z \in \mathcal{Z}_\sigma^0$ can be written uniquely as

$$z = z_\sigma^* + m + k \qquad \text{with } m \in \mathcal{C}_\sigma, \quad k \in K_\sigma^0.$$

Since $\Lambda_\sigma^{\mathrm{dat}}$ and $\Lambda_\sigma^{\mathrm{sh}}$ are affine and K_σ^0 lies in the kernels of their linear parts on V_σ^0, the definitions (13) are well defined and affine.

The set $\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0$ is compact and convex, so its image P_σ^{tr} under the affine map π_σ^{tr} is compact and convex. Because $\mathcal{B}_\sigma^{\mathrm{bal}}$ has nonempty interior and splits exactly as in (7) with $0 \in \operatorname{int} D_\sigma$, the zero-hidden slice carries nonempty relative interior in the ambient body. Therefore P_σ^{tr} has nonempty interior in $\mathcal{C}_\sigma$.

Equation (14) together with (6) gives

$$\left\langle \eta, K_\sigma^{\mathrm{hid}} \eta \right\rangle = \left\langle \eta, \mathcal{K}_\sigma^{\mathrm{bal}} \eta \right\rangle \geq \kappa_\sigma^{\mathrm{hid}} \|\eta\|^2 \qquad (\eta \in H_\sigma),$$

which is (2). The ambient symmetries preserve $\mathcal{Z}_\sigma^0$, V_σ^0, H_σ, K_σ^0, $\mathcal{B}_\sigma^{\mathrm{bal}}$, $\Lambda_\sigma^{\mathrm{dat}}$, $\Lambda_\sigma^{\mathrm{sh}}$, and $\mathcal{K}_\sigma^{\mathrm{bal}}$, so they induce the trace symmetry group and time reversal, preserving P_σ^{tr}, the exact zero-response trace slice, D_σ, and K_σ^{hid}.

For the exact zero-response realization, $\zeta_\sigma(\mathcal{I}_\sigma^{\mathrm{for}})$ is contained in

$$P_\sigma^{\mathrm{tr}} \cap (\Omega_\sigma^{\mathrm{tr,sh}})^{-1}(0)$$

because $\zeta_\sigma^{\mathrm{bal}}(\mathcal{I}_\sigma^{\mathrm{for}}) \subset \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \cap (\Lambda_\sigma^{\mathrm{sh}})^{-1}(0)$. Conversely, if

$$m \in P_\sigma^{\mathrm{tr}} \cap (\Omega_\sigma^{\mathrm{tr,sh}})^{-1}(0),$$

then there exists

$$z \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0$$

with $m = \pi_\sigma^{\mathrm{tr}}(z)$. Since $\Omega_\sigma^{\mathrm{tr,sh}}(m) = \Lambda_\sigma^{\mathrm{sh}}(z) = 0$, we have

$$z \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \cap \left(\Lambda_\sigma^{\mathrm{sh}}\right)^{-1}(0).$$

Because $\zeta_\sigma^{\mathrm{bal}}$ is a diffeomorphism onto that exact shell-zero locus, there is a unique $w \in \mathcal{I}_\sigma^{\mathrm{for}}$ with $z = \zeta_\sigma^{\mathrm{bal}}(w)$, and therefore $m = \zeta_\sigma(w)$. Equation (3) follows directly from (8).

For fixed-data solvability, let $u \in \Omega_\sigma^{\mathrm{tr,dat}}(P_\sigma^{\mathrm{tr}})$. Then $u = \Lambda_\sigma^{\mathrm{dat}}(z)$ for some

$$z \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0.$$

By (9), there exists

$$z_0 \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \cap \left(\Lambda_\sigma^{\mathrm{sh}}\right)^{-1}(0)$$

with $\Lambda_\sigma^{\mathrm{dat}}(z_0) = u$. Setting $m_0 \equiv \pi_\sigma^{\mathrm{tr}}(z_0)$ gives

$$m_0 \in P_\sigma^{\mathrm{tr}} \cap \left(\Omega_\sigma^{\mathrm{tr,sh}}\right)^{-1}(0)$$

and $\Omega_\sigma^{\mathrm{tr,dat}}(m_0) = u$, proving (4).

Finally, let $w \in \mathcal{I}_\sigma^{\mathrm{for}}$ and let $\delta m \in \mathcal{C}_\sigma$ satisfy

$$D\Omega_\sigma^{\mathrm{tr,dat}}\, \delta m = 0.$$

Viewing δm as a vector of V_σ^0, the ambient coercivity inequality (10) applies because $P_\sigma^{\mathrm{co}} \delta m = \delta m$. Hence

$$\|D\Omega_\sigma^{\mathrm{tr,sh}}\, \delta m\|^2 = \|D\Lambda_\sigma^{\mathrm{sh}}\, \delta m\|^2 \geq \kappa_\sigma^{\mathrm{co}} \|\delta m\|^2.$$

This is (5) with $\kappa_\sigma^{\mathrm{tr}} = \kappa_\sigma^{\mathrm{co}}$. □

6. Forced Trace Constituents and Uniqueness

Proposition 2 (All trace-core constituents are forced by ambient dynamics)**.** *Assume Definition 3. Then the following constituents are fixed by the ambient package itself.*

1. *The completion-state space is the faithful quotient of the zero-hidden translation space by the inert kernel, represented canonically by*

$$\mathcal{C}_\sigma = K_\sigma^{0\perp} \cap V_\sigma^0.$$

2. *The completion-state trace body is necessarily the image* (12) *of the zero-hidden ambient body.*

3. *The affine observer-data and shell-response trace maps are necessarily the descended affine readouts* (13) *of the ambient package on the faithful zero-hidden quotient.*

4. *The hidden factor is necessarily the orthogonal complement*

$$H_\sigma = V_\sigma^{0\perp},$$

the hidden body is necessarily D_σ, and the positive hidden operator is necessarily the restriction (14) *of the ambient balance operator.*

5. *The symmetry / time-reversal structure, the exact zero-response trace realization, the fixed-data solvability property, and the trace-kernel coercivity mechanism are all induced by the same ambient package and are not extra trace-level freedom.*

Proof. Items (1) through (4) are exactly the canonical constructions of Definition 4. Once the zero-hidden slice, its inert kernel, the ambient body, and the ambient readouts are fixed, the only faithful completion-state coordinates are those that quotient out the inert kernel, the only trace body compatible with the same ambient body is (12), and the only compatible trace maps are the descended ambient readouts (13). The hidden package is forced by the orthogonal splitting along $V_\sigma^0 \oplus H_\sigma$ together with the balance-operator kernel statement $\ker \mathcal{K}_\sigma^{\mathrm{bal}} = V_\sigma^0$.

Item (5) follows because the ambient symmetry group and time reversal preserve the entire package from which those trace constituents are induced, and Proposition 1 already showed how the exact zero-response realization, fixed-data solvability, and coercivity descend directly to the trace level. □

7. Downstream Shell Coordinates from the Ambient-Induced Trace Core

Definition 5 (Canonical shell coordinates from the trace core)**.** *Assume Definition 2. Define the shell coordinates induced by the trace core as follows.*

1. *Let*
$$L_\sigma^{\mathrm{tr,sh}} : \mathcal{C}_\sigma \to E_\sigma^{\mathrm{amb}}$$
denote the linear part of the affine map $\Omega_\sigma^{\mathrm{tr,sh}}$*. Define*
$$V_\sigma^{\mathrm{ker}} \equiv \ker L_\sigma^{\mathrm{tr,sh}}, \qquad E_\sigma^{\mathrm{sh}} \equiv \operatorname{im} L_\sigma^{\mathrm{tr,sh}}.$$
Let
$$\Pi_\sigma^{\parallel} : \mathcal{C}_\sigma \to V_\sigma^{\mathrm{ker}}$$
be the orthogonal projection onto V_σ^{ker}*, and equip* E_σ^{sh} *with the Euclidean structure transported from* $(V_\sigma^{\mathrm{ker}})^\perp$ *by*
$$L_\sigma^{\mathrm{tr,sh}}\big|_{(V_\sigma^{\mathrm{ker}})^\perp} : (V_\sigma^{\mathrm{ker}})^\perp \xrightarrow{\cong} E_\sigma^{\mathrm{sh}}.$$

2. *Choose the shell offset* $\lambda_\sigma^{\mathrm{sh}} \in E_\sigma^{\mathrm{sh}}$ *by the affine decomposition*
$$\Omega_\sigma^{\mathrm{tr,sh}}(m) = L_\sigma^{\mathrm{tr,sh}}(m) - \lambda_\sigma^{\mathrm{sh}}. \tag{16}$$

3. *The shell-coordinate space is*
$$\mathcal{K}_\sigma \equiv V_\sigma^{\mathrm{ker}} \oplus E_\sigma^{\mathrm{sh}},$$
with product Euclidean structure, and the shell-coordinate isomorphism is
$$T_\sigma(m) \equiv (\Pi_\sigma^{\parallel} m,\, L_\sigma^{\mathrm{tr,sh}}(m)). \tag{17}$$

4. *The shell-coordinate body and shell-data readout are*
$$P_\sigma \equiv T_\sigma(P_\sigma^{\mathrm{tr}}), \qquad \widehat{D}_\sigma \equiv \Omega_\sigma^{\mathrm{tr,dat}} \circ T_\sigma^{-1}. \tag{18}$$

5. *The exact shell affine plane is*

$$\mathcal{A}_\sigma^{\mathrm{sh}} \equiv \left\{(v,e) \in \mathcal{K}_\sigma : e = \lambda_\sigma^{\mathrm{sh}}\right\}, \tag{19}$$

and the exact shell realization is

$$\chi_\sigma(w) \equiv T_\sigma(\zeta_\sigma(w)). \tag{20}$$

6. *The induced symmetry group and time reversal are*

$$\mathbb{G}_\sigma \equiv \left\{\left((g_{\mathrm{st}}|_{V_\sigma^{\mathrm{ker}}},\, g_{\mathrm{sh}}|_{E_\sigma^{\mathrm{sh}}}), g_{\mathrm{hid}}\right) : (g_{\mathrm{st}}, g_{\mathrm{dat}}, g_{\mathrm{sh}}, g_{\mathrm{hid}}) \in \mathbb{K}_\sigma^{\mathrm{tr}}\right\},$$

$$\vartheta_\sigma \equiv \left((\tau_\sigma^{\mathrm{tr}}|_{V_\sigma^{\mathrm{ker}}}, \tau_\sigma^{\mathrm{tr}}|_{E_\sigma^{\mathrm{sh}}}), \tau_\sigma^{\mathrm{tr}}|_{H_\sigma}\right).$$

Proposition 3 (The current exact-shell affine nucleus is already carried by the ambient-induced trace core)**.** *Assume Definitions 2 and 5. Then the shell-coordinate body P_σ, affine shell-data readout $\widehat{D}_\sigma$, exact shell affine plane $\mathcal{A}_\sigma^{\mathrm{sh}}$, exact shell realization χ_σ, hidden body D_σ, hidden operator K_σ^{hid}, induced symmetry / time-reversal structure, fixed-data shell solvability, and affine-kernel coercive control are all determined by the trace core itself. In particular, the shell-coordinate construction yielding the current exact-shell affine nucleus $\mathfrak{N}^{\mathrm{aff}}$ factors through the ambient-induced trace core.*

Proof. Because $L_\sigma^{\mathrm{tr,sh}}|_{(V_\sigma^{\mathrm{ker}})^\perp}$ is a linear isomorphism onto E_σ^{sh}, the map T_σ is a linear isomorphism; with the chosen Euclidean structures it is an isometry. Hence P_σ is compact and convex with nonempty interior whenever P_σ^{tr} is.

If $w \in \mathcal{I}_\sigma^{\mathrm{for}}$, then $\zeta_\sigma(w) \in P_\sigma^{\mathrm{tr}}$ and $\Omega_\sigma^{\mathrm{tr,sh}}(\zeta_\sigma(w)) = 0$. By (16),

$$L_\sigma^{\mathrm{tr,sh}}(\zeta_\sigma(w)) = \lambda_\sigma^{\mathrm{sh}},$$

so $\chi_\sigma(w) \in P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}}$. Conversely, if $q = T_\sigma(m) \in P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}}$, then $m \in P_\sigma^{\mathrm{tr}}$ and the second coordinate condition forces $L_\sigma^{\mathrm{tr,sh}}(m) = \lambda_\sigma^{\mathrm{sh}}$, hence $\Omega_\sigma^{\mathrm{tr,sh}}(m) = 0$. Therefore $m = \zeta_\sigma(w)$ for a unique $w \in \mathcal{I}_\sigma^{\mathrm{for}}$, and so $q = \chi_\sigma(w)$. Thus χ_σ is a diffeomorphism onto the full exact shell slice of P_σ.

Equation (3) immediately gives

$$\widehat{D}_\sigma(\chi_\sigma(w)) = \Omega_\sigma^{\mathrm{tr,dat}}(\zeta_\sigma(w)) = \left(\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)\right).$$

The hidden body, hidden operator, and induced symmetry / time-reversal structure are already part of the trace core.

For shell solvability, let $u \in \widehat{D}_\sigma(P_\sigma)$. Then $u = \Omega_\sigma^{\mathrm{tr,dat}}(m)$ for some $m \in P_\sigma^{\mathrm{tr}}$. By (4), there exists

$$m_0 \in P_\sigma^{\mathrm{tr}} \cap \left(\Omega_\sigma^{\mathrm{tr,sh}}\right)^{-1}(0)$$

with $\Omega_\sigma^{\mathrm{tr,dat}}(m_0) = u$. Hence

$$T_\sigma(m_0) \in P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}}$$

and $\widehat{D}_\sigma(T_\sigma(m_0)) = u$.

Finally, let

$$\delta q \in \mathcal{K}_\sigma$$

satisfy $D\widehat{D}_\sigma\,\delta q = 0$, and let $\delta m \equiv T_\sigma^{-1}(\delta q)$. Since T_σ is an isometry,

$$\|\delta q\| = \|\delta m\|.$$

Moreover,

$$D\widehat{D}_\sigma\,\delta q = D\Omega_\sigma^{\mathrm{tr,dat}}\,\delta m,$$

so the trace-kernel condition gives

$$\left\|D\Omega_\sigma^{\mathrm{tr,sh}}\,\delta m\right\|^2 \geq \kappa_\sigma^{\mathrm{tr}}\|\delta m\|^2.$$

But the second component of δq is exactly $D\Omega_\sigma^{\mathrm{tr,sh}}\,\delta m$. Therefore the shell-normal part of δq obeys the same coercive lower bound, and adjoining any hidden variation $\delta n \in H_\sigma$ gives the same affine-kernel coercive control as in the current nucleus theorem. Thus every downstream shell-coordinate constituent relevant to $\mathfrak{N}^{\mathrm{aff}}$ is already carried by the trace core. □

8. Main Theorems

Theorem 1 (Minimal completion-state shell-trace core necessity / uniqueness from ambient dynamics)**.** *Assume Definition 3. Then:*

1. *the canonical construction of Definition 4 yields a minimal completion-state shell-trace core in the sense of Definition 2;*
2. *every remaining trace-level object singled out by the current trace-core frontier is forced by the ambient package: the completion-state quotient, the compact convex trace body, the affine observer-data and shell-response trace maps, the hidden factor and positive hidden operator, the symmetry / time-reversal structure, the exact zero-response trace realization, the fixed-data solvability property, and the trace-kernel coercivity mechanism;*
3. *any other trace core compatible with the same ambient package differs from the canonical one only by the harmless choice of completion-state origin and by an orthogonal coordinate identification on $\mathcal{C}_\sigma$;*
4. *consequently the shell-coordinate construction yielding the current exact-shell affine nucleus $\mathfrak{N}^{\mathrm{aff}}$ is already carried by the same ambient-induced trace core and does not require the larger completion-core graph presentation as an independently unexplained stopping point.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. For item (3), changing the reference shell point z_σ^* only translates the completion-state coordinate, while changing the faithful orthogonal coordinates on $K_\sigma^{0\perp} \cap V_\sigma^0$ acts only by an orthogonal identification of the same Euclidean completion-state space. No other compatible freedom remains once the ambient zero-hidden slice, inert kernel, ambient body, and ambient readouts are fixed. Item (4) is Proposition 3 together with items (1) and (2). □

Theorem 2 (The derived trace core preserves the same one-metric benchmark package)**.** *Assume Definition 3. Then the benchmark output statements of Definition 1 remain unchanged:*

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi]$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*. Consequently the deeper ambient package, the induced trace core, and the downstream shell-coordinate construction introduce neither a second operative metric nor an enlarged low-energy matter law.*

Proof. The present note changes only the upstream origin of the current live trace datum. It does not alter the primitive shell $\mathcal{S}_{\mathrm{prim}}$, the recorded metric map $\mathscr{G}_{\mathrm{out}}$, or the recorded matter map $\mathscr{T}_{\mathrm{out}}$. Those remain the fixed continuation data of Definition 1, so (1) continues to hold exactly. The induced trace core and the downstream shell-coordinate construction are built to preserve the same exact shell realization of that unchanged primitive shell, not to replace it with a different observer-output law. □

Theorem 3 (The remaining burden moves below the trace core)**.** *Assume the fixed primitive continuation data of Definition 1 and, for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a completion-balance ambient dynamics package in the sense of Definition 3. Then:*

1. *the minimal completion-state shell-trace core of Definition 2 is canonically induced by the ambient package;*
2. *the completion-state quotient, compact convex trace body, affine observer-data and shell-response trace maps, hidden factor and positive hidden operator, symmetry / time-reversal structure, exact zero-response trace realization, fixed-data solvability property, and trace-kernel coercivity mechanism are all forced by that same ambient package;*
3. *the shell-coordinate construction yielding the current exact-shell affine nucleus* $\mathfrak{N}^{\mathrm{aff}}$ *already factors through that ambient-induced trace core;*
4. *the one operative metric remains exactly* $g^{\sharp}$*, the ordinary matter route remains exactly the standard one through that metric, there is no second operative metric, and the low-energy matter law is not enlarged;*
5. *consequently the benchmark-facing burden after the trace-core collapse theorem is discharged in the stronger form now available on the active line: the trace core is no longer an independently unexplained stopping point, but a necessary reduction of a deeper completion-balance ambient dynamics package.*

Proof. Item (1) is Theorem 1(1). Item (2) is Theorem 1(2). Item (3) is Theorem 1(4). Item (4) is Theorem 2. Item (5) is the combined routing consequence. □

Corollary 1 (What remains open after this theorem)**.** *After Theorem 3, the remaining honest burden is no longer to justify the trace core itself on the active primitive-observer-reduction line. The next honest burden is deeper again: derive or justify the completion-balance ambient dynamics package from still more primitive explicit substrate structure, or else prove a sharper inevitability theorem or smaller theorem-level collapse inside that ambient package while preserving the same benchmark one-metric package and claim boundary.*

Proof. Theorem 3 converts the trace core from the current live benchmark-facing burden into a canonical reduction of deeper ambient dynamics. What remains open is why that ambient package itself should hold, rather than becoming the new disciplined stopping point. □

9. Conclusion

The positive result of this note is a direct derivation of the current live internal datum from deeper explicit substrate structure. The previous trace-core collapse theorem had already shown that the full completion-core graph presentation carries no extra benchmark-facing freedom beyond the smaller completion-state shell-trace core. The present note then takes the preferred honest next move left open there: it shows that this smaller trace core is itself canonically induced by completion-balance ambient dynamics.

That gain directly addresses the precise constituents that had become the live burden: the completion-state quotient of the zero-hidden slice, the compact convex trace body, the affine observer-data and shell-response trace maps, the hidden body with positive operator, the induced reversible structure, the exact zero-response trace realization, the fixed-data solvability property, and the trace-kernel coercive control. Within the recorded ambient-dynamics class those are no longer optional inserted ingredients. They are forced outputs, unique up to the harmless affine choice of completion-state origin and orthogonal completion-state coordinates.

The benchmark boundary remains fixed throughout. The same primitive shell remains the shell carrying the observer outputs, so the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. This is therefore a disciplined upstream derivational gain, not a final first-principles completion claim. The note does not yet derive the ambient completion-balance package from ultimate substrate microphysics. That is now the next honest burden. But the trace core itself is no longer the stopping point on the active line.